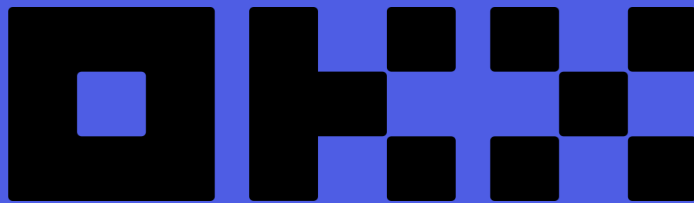


X Layer OP Stack Diff Audit



January 29, 2026

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Summary

Type	Blockchain Infrastructure	Total Issues	17 (10 resolved)
Timeline	From 2026-01-12 To 2026-01-21	Critical Severity Issues	0 (0 resolved)
Languages	Go	High Severity Issues	0 (0 resolved)
		Medium Severity Issues	1 (0 resolved)
		Low Severity Issues	7 (4 resolved)
		Notes & Additional Information	9 (6 resolved)

Scope

OpenZeppelin performed a diff audit of the [okx/optimism](#) repository, comparing commit [23d4cbe](#) on the `dev` branch with Optimism version 1.14.

In scope were the following files:

```
optimism/
├── op-batcher/
│   └── batcher/
│       └── driver.go
├── op-chain-ops/
│   └── cmd/
│       └── withdrawal/
│           └── util.go
├── op-challenger/
│   └── game/
│       └── fault/
│           └── trace/
│               └── super/
│                   └── rollup_configs.go
├── op-conductor/
│   ├── conductor/
│   │   ├── config.go
│   │   └── service.go
│   └── flags/
│       └── flags.go
├── op-node/
│   ├── service.go
│   ├── cmd/
│   │   └── main.go
│   ├── p2p/
│   │   └── discovery.go
│   └── rollup/
│       ├── config_xlayer.go
│       ├── l2_time_xlayer.go
│       ├── superchain.go
│       └── derive/
│           └── span_batch.go
├── op-program/
│   └── chainconfig/
│       └── chaincfg.go
├── op-proposer/
│   ├── flags/
│   │   └── flags.go
│   └── proposer/
│       ├── config.go
│       ├── driver.go
│       └── l2_output_submitter.go
```

```
|      └─ service.go
├─ op-service/
│   └─ crypto/
│       └─ signature.go
│   └─ rpc/
│       ├── handler.go
│       └─ handler_options.go
│   └─ signer/
│       ├── xlayer_adapter.go
│       ├── xlayer_auth.go
│       ├── xlayer_cli.go
│       └─ xlayer_remote_client.go
│   └─ txmgr/
│       ├── cli.go
│       └─ txmgr.go
├─ op-supervisor/
│   └─ supervisor/
│       └─ backend/
│           └─ depset/
│               └─ json.go
```

System Overview

The diff audit focused on modifications to the Optimism OP Stack codebase, including the introduction of a custom remote signing service, hardcoded fork management, L2 timestamp corrections, and additional operational configuration options.

X Layer Remote Signing Service

X Layer replaces local private key storage with an integration to OKX's internal signing infrastructure. Four new files in `op-service/signer/` implement this functionality.

The signing flow works as follows: unsigned transactions are submitted to the remote `/ecologyOperate` endpoint, after which the client polls `/querySignDataByOrderNo` until the signed transaction is returned. Requests are authenticated using an AccessKey/SecretKey pair with AES-encrypted signatures.

The client automatically classifies transactions by component type based on their characteristics: blob transactions are identified as batcher operations, while proposer and challenger operations are detected by their method signatures.

Integration occurs through the transaction manager in `op-service/crypto/signature.go`, where the `SignerFactory` checks for X Layer signer configuration before falling back to other signer types.

Hardcoded Fork Configuration

The file `op-node/rollup/config_xlayer.go` defines hardcoded fork activation times that override values from `rollup.json` or the superchain registry. The current configuration specifies the Jovian fork activation times:

- Mainnet: `December 2, 2025 at 16:00:01 UTC`
- Testnet: `November 28, 2025 at 11:00:00 UTC`

The `ApplyXLayerHardcodedForks` function is invoked at five locations across the codebase (`op-node`, `op-chain-ops`, `op-challenger`, `op-supervisor`) to ensure all components use consistent fork timing.

L2 Timestamp Correction

The `op-node/rollup/l2_time_xlayer.go` file addresses a discrepancy between the L2 genesis timestamps in `rollup.json` and `genesis.json`.

The `GetBatchStartTime` function in span batch derivation applies dual-time logic: batches created before the fork time use the old (incorrect) genesis timestamp for derivation, while batches created after the fork time use the corrected timestamp. This approach preserves backward compatibility with historical batches while correcting the issue for future batches.

Secondary Modifications

The following secondary modifications were identified:

- HTTP Body Limit Configuration: Added the `--rpc.http-body-limit-mb` flag to `op-conductor` for configuring RPC request size limits (default/minimum: 5 MB).
- Proposer Genesis Height: Added the `--genesis-height` flag to `op-proposer` allowing it to skip proposal for blocks up to a specified height.
- Chain Configuration for `op-program`: Added embedded X Layer genesis configurations and gzip decompression support for fault proof execution.

Security Model and Trust Assumptions

The system relies on the remote signature service being safe and trusted. As such, it is assumed that the service correctly implements symmetric signature verification, reliably generates and verifies signatures, does not act maliciously, and adequately protects sensitive information, thereby ensuring the integrity and authenticity of signed requests.

Medium Severity

M-01 Signature Content Collision

The `treeMap` is generated from the request parameters, but it currently only takes into account the parameter values and ignores the parameter keys. These values are then concatenated, which causes different parameter combinations to produce the same signature. For example, the `param1=ab¶m2=c` and `param1=a¶m2=bc` requests both result in the same signature. This behavior can introduce ambiguity and may cause issues when validating requests.

Consider including keys in the signature generation process, for example by concatenating each key with its corresponding value.

Update: Acknowledged, not resolved. The OKX team stated:

From the protocol perspective, the current signature algorithm follows the interface specification determined through negotiation between the remote signature server. The client and the server use the same logic for generating and verifying signatures. Any unilateral algorithm modification will result in signature verification failure, affecting the normal business process.

Therefore, it is impossible to make modifications without coordinating with the upstream server. Secondly, from the perspective of attack feasibility, the `orderId` is filled with the UUID generated by the system, the `projectSymbol` is a fixed value in the configuration file, and all parameters are generated by the code logic rather than accepting external user input. Thus, attackers cannot inject or tamper with the parameter values.

At the same time, the communication link of this interface uses HTTPS encryption transmission and is protected by a dual authentication mechanism of `AccessKey/SecretKey`. The signature key is not exposed externally. Without having the key, attackers cannot forge valid signatures throughout the entire process in the internal network environment. In conclusion, the theoretical parameter collision scenario does not have practical usability in the current business context.

Low Severity

L-01 Incorrect Log Level Used in X Layer Fork Configuration

The `ApplyXLayerHardcodedForks` function uses `log.Error` for expected scenarios: when `no rollup configuration is provided` and when `processing non-X Layer chains`. Both cases represent normal operation rather than error conditions. Using error-level logging for routine scenarios can introduce noise into monitoring systems and may trigger false alerts, making it more difficult to identify genuine errors.

Consider changing these log statements to `log.Debug` or `log.Info`, as returning unchanged configuration for non-X Layer chains is the expected behavior that does not warrant error-level logging.

Update: Resolved in [pull request #194](#) at commit [4789bc9](#).

L-02 Lack of Input Validation Leads to Integer Overflow Risk

The `GetBatchStartTime` function in `l2_time_xlayer.go` is responsible for calculating the effective start time of an L2 batch. This calculation involves adding a `relTimestamp` (relative timestamp) to a `genesisTimestamp` or a chain-specific base time. This timestamp derivation is a fundamental operation that underpins the L2 chain's time progression and fork logic.

The `relTimestamp` parameter, which is a `uint64`, is used in arithmetic additions without any preceding validation or bounds checking. If an excessively large value is provided for `relTimestamp`, the addition operation (e.g., `genesisTimestamp + relTimestamp`) can result in an integer overflow. In Go, `uint64` arithmetic silently wraps around upon overflow, resulting in an incorrect and unexpectedly small `batchStartTime`. This erroneous timestamp could cause the fork logic to evaluate incorrectly, misapplying historical or fixed time rules, or producing timestamps that are fundamentally out of sequence, thereby risking L2 state integrity and potentially leading to consensus failures.

Consider implementing comprehensive validation for `relTimestamp` to ensure that it remains within a realistic and expected range. Furthermore, for all critical arithmetic operations

involving timestamps, consider incorporating explicit overflow checks or utilizing a safe math library to prevent silent integer wraparound, ensuring that any potential overflow condition is detected and handled gracefully.

Update: Acknowledged, not resolved. The OKX team stated:

relTimestamp is written by the centralized op-batcher. Therefore, under normal circumstances, the probability of addition overflow is extremely low. Even if an overflow does occur, the batch will be dropped for being invalid, which is part of the normal batch handling process, and there is no need to handle the overflow issue separately. Furthermore, the official code does not handle overflow either, so here we keep it consistent with the official implementation.

L-03 File System Write Without Validation

The `saveFixedRollupJSON` function is a utility responsible for writing a modified rollup configuration object to a file present in the local filesystem. The function takes a `rollupConfigPath` parameter to determine the file's location and creates it with `0644` permissions.

The implementation does not follow defensive programming best practices. The `rollupConfigPath` is used directly without validation or sanitization. While the path is not expected to be controlled by an untrusted actor in its current context, this lack of input validation could pose a risk if the function's usage changes in the future. Additionally, the `0644` file permissions make the file world-readable, which may be overly permissive for a configuration file that could potentially store sensitive information as the system evolves. To improve the robustness and long-term security of this function, proactive hardening is recommended.

Consider applying path sanitization and validation to ensure the `rollupConfigPath` is well-formed and resides within an expected directory. Furthermore, to better align with the principle of least privilege, consider using more restrictive default file permissions, such as `0600`, to protect the file's contents from unintended access.

Update: Resolved in [pull request #193](#) at commit [85e083d](#). The OKX team stated:

We have chosen to preserve the existing file permissions instead of forcing 0600.

Why We Are Not Fully Adopting the Audit Recommendations:

Path validation: The file is already opened in `NewRollupConfig()` before this function is called, so the path is trusted. The codebase uses paths directly without validation, so adding it here would be inconsistent.

File permissions: The file already exists with user-configured permissions. Preserving them respects user intent and avoids unintended changes. If stricter permissions are needed, they should be set at file creation, not during modification.

This approach maintains consistency with the existing codebase while preserving user-configured security settings.

L-04 Infinite Polling Loop Without Maximum Timeout

The `waitSignResult` function in `xlayer_remote_client.go` uses an unbounded polling loop that lacks a maximum timeout or retry limit. The function only exits upon context cancellation or a successful response, allowing it to poll indefinitely if the remote signer returns empty results without errors.

In a degraded signer scenario, each pending request continuously polls at one-second intervals while holding a goroutine and internal locks. Multiple such requests can quickly exhaust memory, overload the signer with repeated network calls, and block other signing operations due to mutex contention. This behavior can lead to a denial of service for batch submissions.

Consider adding a hard timeout, a maximum poll attempt limit, and clearer error handling to ensure that bounded resource usage and more predictable failure behavior.

Update: Resolved in [pull request #192](#) at commit [5a39d92](#).

L-05 Empty `Close` Function

The `xlayer_remote_client` implements a `Close` function, but it currently contains no logic to release any allocated resources.

Consider adding logic to properly release resources, such as closing connections, freeing memory, or cleaning up other allocated resources, to prevent potential leaks.

Update: Resolved in [pull request #190](#) at commit [dd3bb09](#).

L-06 Use of ECB Mode for AES Encryption

The `genAuth` function constructs a `treeMap` from HTTP request parameters, and the `generateSignature` function `hashes` this map and then `encrypts` the resulting hash using `AES in ECB mode`. AES in ECB mode is widely considered insecure, as it does not provide semantic security and can reveal patterns in the input. Moreover, encrypting a hash in this manner is conceptually unnecessary and does not provide the security properties typically required for request authentication. The overall design appears to be a custom and non-standard attempt to implement message authentication, which increases the risk of cryptographic weaknesses and implementation errors.

Consider replacing the custom hash-and-encrypt logic with a standard HMAC-SHA256 over the request parameters using a shared secret key. This would entirely eliminate the need for AES while providing a simpler and more secure authentication mechanism.

Update: Acknowledged, not resolved. The OKX team stated:

The current signature algorithm follows the interface authentication specification of the remote signature server. The client and the server use the same signature generation and verification logic. Any unilateral algorithm change will result in signature verification failure, affecting the normal business process. Therefore, it cannot be changed.

From the perspective of actual risks, although the AES ECB mode theoretically has the risk of mode leakage, in the current scenario:

- 1. the encrypted content is a SHA256 hash value, which is already high-entropy random data and does not have exploitable repetitive patterns*
- 2. this signature is used for request authentication between internal systems rather than data encryption, the key is not exposed externally and the communication link is already protected by HTTPS*
- 3. the attacker cannot infer valuable business information by observing the ciphertext. And it is all an internal network environment and does not have attackability*

L-07 Insufficient Verification of Remotely Signed Legacy Transactions

`XLayerRemoteClient` sends transaction signing requests to a remote service and, upon receiving a response, attempts to verify its integrity via the `verifySignedTransaction` function. This function checks that the parameters of the signed transaction match those of the

original request. A special case within this logic exists in `verifyLegacyTxFields` to handle scenarios where the remote signer converts an EIP-1559 (Type 2) transaction into a legacy (Type 0) transaction.

The verification logic for this transaction type conversion is overly permissive. The function `checks` whether the resulting legacy `gasPrice` falls within a broad $\pm 20\%$ tolerance of the original EIP-1559 `gasFeeCap`. If this check fails, the function only logs a warning and proceeds as though verification succeeded. This behavior allows a faulty remote signer to return a transaction with a gas price that differs significantly from the intended value. As a result, the client may unknowingly broadcast a transaction that either pays an excessive fee or uses a gas price too low to be included in a block.

To ensure that transaction parameters are not unexpectedly modified, the verification logic should be enforced more strictly. Consider reducing the tolerance for `gasPrice` deviation to a much smaller, clearly defined threshold. In addition, the function should return an error upon verification failure rather than merely logging a warning, thereby allowing the calling service to handle invalid transactions appropriately.

Update: Acknowledged, not resolved. The OKX team stated:

After a business assessment, it was decided to maintain the current design. Firstly, the remote signature service is a trusted service independently developed and deployed within the internal network environment. There is no risk of external attacks or malicious tampering with transaction parameters. Secondly, the current design prioritizes business continuity - in the blockchain scenario, a failed transaction signature means that batch submission is interrupted, the proposer is unable to generate blocks, or the challenger is unable to respond. These business impacts are much greater than a minor deviation in gas price.

The $\pm 20\%$ tolerance design is to accommodate reasonable calculation differences that the remote signature service may produce in specific scenarios (such as the transition from EIP-1559 to Legacy), while retaining observability through warning logs. The operation team can monitor the logs to detect anomalies in a timely manner. If the verification failure is changed to a direct error report, it may cause abnormal transaction processes to be blocked due to incorrect judgment of boundary conditions, which is contrary to the core business goal of "ensuring the availability of chain operations". In summary, the current design achieves a reasonable balance between business availability and security monitoring in a trusted internal network environment.

Notes & Additional Information

N-01 Misleading Comment Regarding Blob Transaction Assembly

Within the `xlayer_remote_client.go` file, the `postSignRequestAndWaitResult` function handles responses from the remote signing service. After receiving and unmarshalling the signed transaction, this function contains specific logic to manage EIP-4844 blob transactions, which consist of a transaction payload and an associated "sidecar" containing the blob data.

The remote signing service operates only on the transaction payload and does not receive, sign, or return the blob sidecar. Consequently, the client is responsible for re-attaching the original sidecar to the signed transaction payload to construct a complete and valid E.g., EIP-4844 blob transaction. While this re-attachment logic is correctly and necessarily implemented, a code comment immediately preceding this operation states, "[Do not reassemble!](#)". This comment is incorrect and misleading, as re-assembly is a required step in this workflow. This discrepancy creates confusion regarding the code's intent and the signing service's operational contract.

To ensure the code is clear and maintainable, the inaccurate comment should be removed. Consider replacing it with a new comment that explains why the sidecar re-attachment is necessary, clarifying that the remote signing service does not handle the sidecar and that the client must therefore re-assemble the complete blob transaction after the payload is signed.

Update: Resolved in [pull request #190](#) at commit [dd3bb09](#).

N-02 Sensitive Credentials Can Be Exposed via CLI Arguments

The X Layer remote signer client is configured through flags defined in the `xlayer_cli.go` file. These flags include `xlayer-signer.access-key` and `xlayer-signer.secret-key`, which allow users to provide credentials for authenticating with the remote signing service directly via the CLI.

Passing secrets such as access keys and secret keys as command-line arguments is an insecure practice. On multi-user operating systems, the arguments of a running process are often visible to other users on the same machine who can inspect the system's process list. This unintentional exposure can lead to the leakage of sensitive credentials, which may be captured by other processes or users, compromising the security of the signing service.

To mitigate this vulnerability, consider removing support for supplying these credentials via CLI arguments. Instead, rely on more secure mechanisms for secret management. This could include exclusively using environment variables for credential injection or implementing support for reading credentials from a protected configuration file with strict permissions.

Update: *Acknowledged, not resolved. The OKX team stated:*

We support both the environment variable and startup command methods for starting. Moreover, the test network and the main network are accessible only to specific release machines and personnel. Others are not allowed to enter the deployment system.

N-03 Missing Block Ordering Validation

The batcher's block loading loop calculates the gap between unsafe and safe L2 blocks by [performing unsigned integer subtraction](#). The unsafe L2 block number represents the absolute tip of the L2 chain, while the safe L2 block number represents blocks derived from L1 data. By design, the `curL2SafeBlock.Number ≤ curL2UnsafeBlock.Number` invariant should always hold.

However, the code does not validate this invariant before performing the subtraction. If the safe block number were to exceed the unsafe block number—due to a bug in the node implementation, database corruption, or a transient inconsistency during reorg handling—the unsigned subtraction would wrap around to produce an extremely large value. This would result in misleading log output that could mask the underlying critical issue, making it difficult to detect and diagnose serious synchronization problems or state corruption.

Consider adding explicit validation to detect and handle this invariant violation before calculating the gap. Such a check should log an error with detailed block information when the safe block number exceeds the unsafe block number, as this condition indicates a critical system failure.

Update: *Resolved in [pull request #196](#) at commit [0fe9396](#).*

N-04 Missing Upper Bound Validation for HTTP Body Limit Parameter

The HTTP body limit parameter specified by `rpc.http-body-limit-mb` allows operators to configure the maximum HTTP request body size in megabytes. While the [code enforces](#) a minimum value of 5 MB, no maximum value is defined, allowing operators to configure excessively large limits. This could lead to server memory exhaustion and potentially result in a denial-of-service condition.

Consider adding an upper-bound validation to restrict the maximum allowable value for this parameter.

Update: Resolved in [pull request #195](#) at commit [e22b117](#).

N-05 Gzip Decompression Without Size Limits

The [gzip decompression logic](#) for genesis files does not enforce any limits on the size of the decompressed output. This creates a risk that a specially crafted compressed genesis file could expand to an excessive size during decompression. As a result, the system may experience uncontrolled memory consumption, potentially leading to memory exhaustion and a denial-of-service condition.

Consider introducing explicit bounds on the amount of data that can be decompressed, for example by using a size-limiting reader to restrict the maximum allowable output.

Update: Resolved in [pull request #197](#) at commit [7c6d8e8](#).

N-06 Timeout Parse Error Silent Fallback

The system [silently falls back to a default value](#) when an invalid timeout configuration is provided. This behavior can obscure configuration errors and make them difficult to detect during deployment or operation. As a result, operators may believe that a custom timeout is in effect when it is not, potentially leading to unexpected runtime behavior.

Consider returning an explicit error or, at minimum, logging a clear warning when an invalid timeout configuration is encountered.

Update: Acknowledged, not resolved. The OKX team stated:

The remote signature configuration is not accessible to the public. Only specific publishing machines and specific professionals can configure this parameter. So, the risk is almost zero.

N-07 Standardize Error Messages in `xlayer_remote_client`

The [implementation in `xlayer\_remote\_client`](#) mixes Chinese and English in error messages. Mixing languages can lead to encoding issues in some logging systems and makes the codebase harder to maintain.

Consider standardizing error codes from the remote API instead of relying on string matching for error handling. This approach would improve maintainability and reduce the risk of logging or encoding-related issues.

Update: Resolved in [pull request #192](#) at commit [b684dc0](#).

N-08 UUID Generation Error Not Handled

Within `xlayer_remote_client.go`, the `RefOrderID` is generated using `uuid.New()`. While `uuid.New()` is designed to never fail under normal conditions, it could panic in extreme edge cases, such as entropy exhaustion.

Consider adding logic to handle a potential panic from `uuid.New()` to help ensure the application remains resilient under unexpected conditions.

Update: Resolved in [pull request #192](#) at commit [5a39d92](#).

N-09 The AES Key Length is Not Validated

If `SecretKey` has an invalid length, `aes.NewCipher` will return an error. Currently, there is no pre-validation of the key length, which allows invalid configurations to propagate further. In addition, error messages from the crypto library may leak sensitive information.

Consider validating key length at configuration time.

Update: Acknowledged, not resolved. The OKX team stated:

This configuration can only be set by specific release machines and specific personnel after approval. It is not accessible to the public and there will be no issues of configuration errors.

Conclusion

The diff audit thoroughly examined the recent changes to the Optimism OP Stack, highlighting updates to remote signing, fork management, L2 timestamp handling, and operational configurations. The audit identified one medium-severity issue related to signature collision, along with several minor issues, resulting in recommendations aimed at improving code consistency, robustness, and readability.

The OKX team is appreciated for their excellent responsiveness and collaboration, with timely and clear explanations that contributed positively to the overall audit process.

Appendix

Issue Classification

OpenZeppelin classifies smart contract vulnerabilities on a 5-level scale:

- Critical
- High
- Medium
- Low
- Note/Information

Critical Severity

This classification is applied when the issue's impact is catastrophic, threatening extensive damage to the client's reputation and/or causing severe financial loss to the client or users. The likelihood of exploitation can be high, warranting a swift response. Critical issues typically involve significant risks such as the permanent loss or locking of a large volume of users' sensitive assets or the failure of core system functionalities without viable mitigations. These issues demand immediate attention due to their potential to compromise system integrity or user trust significantly.

High Severity

These issues are characterized by the potential to substantially impact the client's reputation and/or result in considerable financial losses. The likelihood of exploitation is significant, warranting a swift response. Such issues might include temporary loss or locking of a significant number of users' sensitive assets or disruptions to critical system functionalities, albeit with potential, yet limited, mitigations available. The emphasis is on the significant but not always catastrophic effects on system operation or asset security, necessitating prompt and effective remediation.

Medium Severity

Issues classified as being of medium severity can lead to a noticeable negative impact on the client's reputation and/or moderate financial losses. Such issues, if left unattended, have a moderate likelihood of being exploited or may cause unwanted side effects in the system.

These issues are typically confined to a smaller subset of users' sensitive assets or might involve deviations from the specified system design that, while not directly financial in nature, compromise system integrity or user experience. The focus here is on issues that pose a real but contained risk, warranting timely attention to prevent escalation.

Low Severity

Low-severity issues are those that have a low impact on the client's operations and/or reputation. These issues may represent minor risks or inefficiencies to the client's specific business model. They are identified as areas for improvement that, while not urgent, could enhance the security and quality of the codebase if addressed.

Notes & Additional Information Severity

This category is reserved for issues that, despite having a minimal impact, are still important to resolve. Addressing these issues contributes to the overall security posture and code quality improvement but does not require immediate action. It reflects a commitment to maintaining high standards and continuous improvement, even in areas that do not pose immediate risks.